

OMANSHU BHATT

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omanshu09.github.io/omanshu-repository

PROFESSIONAL SUMMARY

AI/ML and Full-Stack Engineer building production-grade retrieval and reasoning systems. I architect end-to-end solutions, from distributed data pipelines to low-latency APIs that bridge experimental AI with practical applications. My work includes a research-driven web evidence search engine, RAG-based code intelligence platform, where I tackle challenges in hybrid search, distributed indexing and system reliability. I don't just experiment; I build and ship systems that reduce technical complexity through clean and practical engineering.

TECHNICAL SKILLS

Languages: Python, Go, Java, JavaScript, C

Web & Frameworks: React.js, FastAPI, Node.js, Django, HTML, CSS

Data & Databases: Elasticsearch, PostgreSQL, Redis

AI / ML: PyTorch, LangChain, Generative AI, RAG, Vector Databases

Cloud, DevOps & Systems: AWS, Docker, Kubernetes, Kafka, CI/CD (Jenkins, GitHub Actions)

Core CS: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming

PROJECTS

Web Research & Evidence Engine <https://search-engine-frontend-xi.vercel.app/>

- Engineered a research-driven web evidence search engine that performs query processing, multi-source web retrieval, content extraction, relevance filtering, ranking to identify high-value sources and aggregates retrieved evidence to detect agreement and contradictions before generating structured, source-backed responses through LLM-based evidence synthesis.
- Architected the end-to-end application with Python/FastAPI services for search orchestration, retrieval, ranking and evidence analysis, a React-based research interface and Dockerized deployment, integrating modular retrieval and processing components into a reliable pipeline for automated web research.

Tech: Web Search, Information Retrieval, Evidence Extraction, RAG, LLM, FastAPI, React, Docker

Benchmarked across 100+ research queries, achieving ~85% relevant source retrieval with sub-10s median response time.

Codebase Intelligence Platform

- Built an AI-powered platform that ingests GitHub repositories and enables natural-language querying across source code. Implemented a RAG pipeline with code embeddings and vector search to retrieve relevant files and functions before generating contextual LLM responses.
- Developed FastAPI services for repository ingestion, processing, retrieval, and inference, with Docker-based deployment.

Tech: Python, FastAPI, React, RAG, LLM, Vector Database, Docker

Processed over 1,500 source code files across multiple repositories, enabling natural-language retrieval with ~75% relevant file recall.

EXPERIENCE

Trainee Intern | Indian Oil Corporation Limited (IOCL)

- Collaborated with cross-functional teams on technical documentation and process optimization.
- Supported data analysis initiatives for operational efficiency improvements, and gained exposure to enterprise-level engineering workflows and quality-assurance practices.

Virtual Intern — AI/ML & Robotics | Kodacy

- Completed two virtual internship tracks covering AI/ML fundamentals, data analysis, and robotics/automation concepts.
- Developed robotics simulation prototypes using Python and relevant frameworks.

CERTIFICATIONS

- Oracle — Agentic AI Certified Foundations Associate
- TCS iON Career Edge - Young professional
- NASA — Hyperspectral Data for Land and Coastal Systems
- Deloitte — Cyber Job Simulation
- Commonwealth Bank — Software Engineering Job Simulation
- University of Oxford — AI, Justice and rule of law

EDUCATION

Bachelor of Technology (B.Tech), Computer Engineering

Govind Ballabh Pant University of Agriculture and Technology, Pantnagar | 2022 – 2026 | CGPA: 7.2

Beersheba Senior Secondary School, Almora

Class XII — 90.4%

Class X — 95.3%